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# EXPERIMENT – 11

## Performing IP Spoofing Using a GitHub Tool in Kali Linux (Educational Demonstration)

### Aim

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

### Software Requirements

**OperatingSystem:** Windows / Linux / macOS
**VirtualizationSoftware:** Oracle VirtualBox
**GuestOperatingSystem:** Kali Linux
**ToolSource:** GitHub (IP Spoofing Tool – Educational Use)
**ProgrammingLanguage:** Python / Bash (as per tool)
**SystemRequirements:**

- RAM: Minimum 4 GB (8 GB recommended)
- Disk Space: Minimum 30 GB
- Processor: Intel/AMD with Virtualization support

### Introduction

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates

### Procedure

### Step 1: Start Kali Linux Virtual Machine

1. OpenOracleVirtualBox. 2. StartKaliLinux.

### Step 2: Open Terminal

1. LaunchTerminalinKaliLinux.
2. Updatesystempackages:sudoaptupdate

### Step 3: Clone IP Spoofing Tool from GitHub

1. Clonetherepository:
2. gitclonehttps://github.com/<username>/<ip-spoofing-tool>
3. Navigatetothetooldirectory:cdip-spoofing-tool

**Step 4: Install Dependencies**

1. Install required dependencies: sudo apt install python3
2. pip3 install -r requirements.txt

**Step 5: Configure the Tool**

1. Openthescriptfile:nanospoof.py
2. Configure:
   - o TargetIPaddress
   - o SpoofedsourceIPaddress

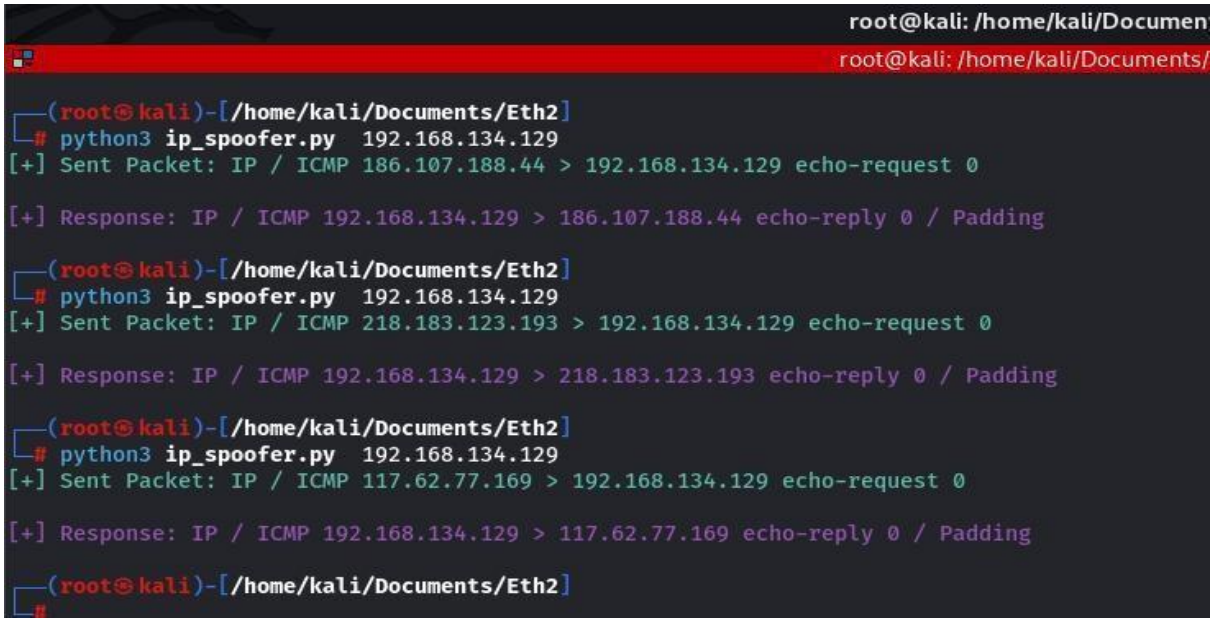
**Step 6: Execute IP Spoofing**

1. Runthetoolwithrootprivileges:sudopython3spoof.py
2. ObservepackettransmissionwithspoofedIPaddresses.

**Step 7: Monitor Network Traffic (Optional)**

1. UseWiresharkortcpdump:sudotcpdump
2. VerifyspoofedsourceIPpackets.

**Output**




**Result**

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux.
The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.